

“THIN-FILM SOLAR CELLS OFFER A LOWER-COST SOLUTION”

With the race on to have the most efficient, low-cost solar module on the market, GE is focusing its R&D efforts on thin-film photovoltaic (PV) technology in conjunction with PrimeStar Solar, which is headquartered in Arvada, Colorado, USA.

Mano Manoharan (Mano), general manager, GE Global Research, based out of GE's largest multidisciplinary center in the world at GE John F. Welch Technology Center, Bengaluru, has immense technical expertise and experience in the areas of nanotechnology, fracture mechanics, surface engineering and mechanical properties of materials. He speaks to Uma Bansal, deputy editor at EFY, about the trends in solar cells and the reason for choosing cadmium-telluride (CdTe) technology for their modules



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Of late, what have been the major advances in solar cell technology?

Current advancements are in the areas of overall efficiency improvements of thin-film modules and bringing their costs down. Advancements are also being made in the direction of maintaining high reliability and expected lifetime of 20 years.

What areas of thin-film solar cells are being researched worldwide and at GE?

We have a global team of all our four of GE Global Research centres working on thin-film solar. In addition, GE scientists are working closely with our partners at PrimeStar to develop our thin-film technology.

We are looking at several key areas in order to achieve a best-in-class product, including device efficiency, reliability, production and installation costs, and manufacturability. Hundreds of technologists in Germany, China, India and the United States are working on GE solar technologies today, addressing these challenges.

The market-leading thin-film modules today are slightly over 11 per cent efficient with costs of around 84 cents/watt.

The team in Munich is utilising world-class indoor and outdoor solar module system test facilities where they study finished module performance to identify and address degradation mechanisms and packaging issues. In addition, this team also brings deep expertise at the system level. This allows them to define system-level optimised features and metrics for the module.

In China, where most of the world's CdTe raw materials are found, researchers at GE's China Technology Center in Shanghai are focused on CdTe materials and the impact they have on device performance. Improving material quality and developing advanced materials characterisation techniques are key topics being addressed by the Shanghai team.

In India, GE is leveraging extensive modeling capabilities at its GE John F. Welch Technology Center in Bengaluru. Unlike the exclusively experimental approach favoured by many in this field, GE believes that dramatic improvements